

**Project Manual
for
CRIP Headgate Rock Dam
Stop Log Lifting Beam End Guide Modifications**



**Bureau of Indian Affairs
Office of Trust Services – Division of Water and Power**

Prepared By:



Project No. 4626.12494.01

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March, 6, 2026

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PART 1 - GENERAL

1.01 SECTION INCLUDES

- A. For each bid item, the work will be measured and paid for on either a unit price basis or on a lump sum basis. The quantities provided in the descriptions of the bid items are estimates of the actual quantities of work only and are included solely for guidance in determining the probable cost of work. The actual quantities of work may differ from the quantity estimates provided. The quantities for measurement and payment for unit price bid items shall be as calculated to the neatlines shown on the drawings. All labor, equipment, materials, and incidentals required to complete the work shall be considered subsidiary to that bid item and will not be paid for separately. The contractor agrees to make no claim for damages, loss of anticipated profits, or otherwise, due to differences between the actual work quantities and the quantity estimates provided.
- B. Payment will be made only for each item included in the bid. No work will be paid for that is not completed in accordance with the drawings and specifications and accepted by the BIA Representative. No labor, equipment, materials, or incidentals required to complete the work will be furnished by the BIA Representative.
- C. The basis of measurement and payment for each bid item is described below. A general listing of bid items, accompanied by a brief summary of the work, is also provided. The summaries are not intended to completely describe all work. Refer to the drawings and specifications for detailed information on each bid item.

1.02 ACRONYMS

- A. Not applicable

1.03 MEASUREMENT AND PAYMENT PROCEDURES

Item 101: Lifting Beam End Guide Modifications and Transport

- a. This item consists of transporting the Owner-furnished lifting beam to the fabrication shop, modifying the lifting beam, and delivering the modified lifting beam to the project site.
- b. This bid item includes, but is not limited to:
 - i. Transporting the lifting beam from the project site to the fabrication shop and transporting the lifting beam back to the project site after the modifications are complete.
 - ii. Furnishing all labor, equipment, and materials to modify the lifting beam as described in the drawings and specifications.
 - iii. Developing, compiling, and submitting all information described under the in Section 01 33 00 – Submittal Procedures.
 - iv. Related submittal reviews, resubmittals, and coordination required to achieve final, approved/accepted submittals.

- c. Measurement and payment will be at the Contract lump sum price. Contractor may request payment on this item after the shop inspection and delivery to the project site.

Item 102: Facilitate Final Test

- a. This item consists of the Contractor's attendance and facilitation of the Final Test at the project site.
- b. This bid item includes, but is not limited to:
 - i. Coordinating with BIA to select a date and time for the Final Test.
 - ii. Furnishing a crane and crane operator to support BIA's field-directed testing.
 - iii. Attendance by Contractor's representative at the Final Test and demonstration by this individual of the adjustable end guide functionality. End guide adjustments by the Contractor may be required based on the outcome of the trial stop log installation test.
- c. Measurement and payment will be at the Contract lump sum price. BIA will pay the Contractor when the work associated with this item is complete.

PART 2 - PRODUCTS

Not Used

PART 3 - EXECUTION

Not Used

END OF SECTION

PART 1 - GENERAL

1.01 SECTION INCLUDES

- A. This section covers the requirements for submittals by the Contractor of descriptive information that will enable a determination of whether the Contractor's proposed materials, equipment, and work methods are in general conformance to the design concept and in accordance with the specifications.

1.02 REQUIRED SUBMITTALS

- A. Consists of:
1. Permits
 2. Drawings
 3. Product Data
 4. Certifications
 5. Manuals, Shop Drawings, and Specialized Tools
 6. Samples
 7. Test Reports

1.03 DEFINITIONS

- A. Days: Calendar days.
- B. Required Submittal Number (RSN): Identifies items to be submitted together as a complete submittal.
- C. Submittal Types, as listed in Table 01 33 00A – List of Submittals:
1. A – Approval:
 - a. BIA Representative will accept or reject Approval submittals.
 2. I – Informational:
 - a. The BIA Representative will acknowledge receipt and acceptance of these submittals.
 - b. The BIA Representative may request additional clarification, information, and/or documentation on the content of an informational submittal. If the BIA Representative arrives at different conclusions than described by the informational submittal, the BIA Representative may provide recommendations for Contractor's consideration.

1.04 SUBMITTAL REQUIREMENTS

- A. In case of conflict between requirements of this section and requirements included elsewhere in these specifications, requirements included elsewhere take precedence.

B. Submittal Letter of Transmittal:

1. Contract number.
2. Project name.
3. RSN.
4. List of documents/content included with the submittal.
5. Relevant discussion and/or information that the BIA should be aware of to effectively review and fully understand the submittal content.
6. Statement that the submittal has been reviewed and approved by the Contractor's authorized representative.
7. Signature of Contractor's authorized representative.

1.05 SUBMITTAL CONTENT

- A. Provide a complete submittal package: Include all documentation and information necessary to satisfy the submittal requirements. If an incomplete submittal package is submitted for review, identify what elements are missing and the reason those elements were omitted. The BIA Representative reserves the right to reject partial submittals.
- B. Product Data Sheets: For manufacturer data with multiple items, the elements being submitted for review shall be clearly identified.
- C. Certifications:
1. Certifications by a registered professional: Signed and sealed by registered professional.
 2. Manufacturer's certifications: Signed by authorized representative of manufacturer.

1.06 SUBMITTALS PROCEDURES

- A. Contractor shall review and approve all submittals prior to submitting them. Submittals without evidence of Contractor's approval will be returned for resubmission.
- B. Submittals shall be submitted electronically in portable document file (pdf) format through a web-based portal. Submittals that cannot be submitted electronically, such as material/product samples, shall be submitted to the BIA Representative. Note that in the case of product samples, four samples shall be submitted.
- C. Distribution and review status of individual submittals shall be administered through the web-based portal.
- D. Reviewed submittals and submittal disposition shall be returned/sent to the Contractor electronically in pdf format.

1.07 REVIEW OF SUBMITTALS

A. Time Required:

1. Submittal review will require 28 days for review of each submittal or resubmittal, unless otherwise specified.

2. The review period for a submittal (or resubmittal) begins when a complete submittal package is received.

B. Return of Submittals:

1. Submittals will be returned either approved, approved as noted, not approved, or acknowledging receipt.
2. For submittals not approved:
 - a. Revise and resubmit for approval.
 - b. Show changes and revisions.
 - c. Resubmit returned submittals within 28 days after receiving the comments.
 - d. In the resubmittal transmittal letter, describe how the submittal was revised and the how comments on the initial (or previous) submission were addressed.

PART 2 - PRODUCTS

Not Used

PART 3 - EXECUTION

Not Used

Table 01 33 00A – List of Submittals

Submittal No.	Section Title	Submittals Required	When to Submit	Type
05 05 13-1	Painting and Coatings	Product Data	Within 30 days of Notice of Award	A
05 05 13-2	Painting and Coatings	Shop Testing and Inspection Plan	Within 30 days of Notice of Award	A
05 05 13-3	Painting and Coatings	Shop Testing and Inspection Results	No more than 14 days after completing the test or inspection	I
40 05 59.16-1	Steel Stoplogs	Shop Drawings and Product Data	A minimum of 45 days before beginning fabrication.	A
40 05 59.16-2	Steel Stoplogs	Shop Testing and Inspection Plan	Within 30 days of Notice of Award	A
40 05 59.16-3	Steel Stoplogs	Shop Testing and Inspection Results	No more than 14 days after completing the test or inspection	I

END OF SECTION

PART 1 - GENERAL

1.01 DESCRIPTION

- A. This specification section includes requirements for surface preparation, painting, and paint testing of all ferrous metals to be supplied per the contract documents.
- B. *BIA is adopting this section from the original fabrication specifications. Changes from the original specification are indicated herein by strikeout (for deletions) and italics (for additions)."*

1.02 REFERENCE SPECIFICATIONS, CODES, AND STANDARDS

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only. The most recent editions of the publications shall govern.
1. American Society for Testing and Materials
 - a. A380 Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems.
 - b. D2369 03 Test Method for Volatile Content of Coatings
 - c. D2697 Test Method for Volume Non-Volatile Matter in Clear or Pigmented Coatings.
 - d. D2832 92(R1999) Guide for Determining Volatile and Nonvolatile Content of Paint and Related Coatings
 - e. D3359 Standard Test Methods for Measuring Adhesion by Tape Test
 - f. D4541 Pull Off Adhesion
 - g. D4285 Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel
 - ~~h. D5163 Standard Practice for Discontinuity (Holiday) Testing of Non-Conductive Protective Coating on Metal Substrates~~
 - ~~2. National Association of Corrosion Engineers International (NACE) RP 0188-99. Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates.~~
 3. The Society For Protective Coatings (SSPC-Formerly The Steel Structure Painting Council)
 - a. SP 1 Surface Preparation Specification No. 1, Solvent Cleaning
 - b. SP 2 Hand Tool Cleaning
 - c. SP 3 Power Tool Cleaning
 - d. SP 5 White Metal Blast Cleaning
 - e. SP 6 Commercial Blast Cleaning

- f. SP 7 Brush-Off Blast Cleaning
- g. SP 10 Near-White Blast Cleaning
- h. SP 11 Power Tool Cleaning to Bare Metal
- i. SP 12 High Pressure Water Jetting
- j. Guide No.3 (PA 3) Guide to Safety in Paint Applications
- k. Specification No.2 (PA2) Measurement of Dry Film Thickness with Magnetic Gauges

1.03 SUBMITTALS

Submit the following in accordance with the requirements of Section 01 33 00 – Submittals:

- A. Product manufacturer's technical data sheets showing the following information:
 - 1. Descriptive data.
 - 2. Performance characteristics.
 - 3. Product data sheets.
 - 4. Surface preparation.
 - 5. Application instructions.
 - 6. Patching, repairs, and touch-up instructions.
 - 7. Material safety data sheets.
 - 8. Paint and color schedule.
 - 9. ~~Color swatches.~~
- B. Inspection reports and data collection sheets for surface preparation and profile checks, coating thickness measurements for each coat, ~~holiday tests~~, ambient conditions, and other pertinent records.
- C. QA/QC Procedure ~~Manual~~ and Inspection and Testing Plan for review and acceptance.
- D. ~~Manufacturer's product warranties.~~
- E. ~~Certificate of Conformance.~~
- F. ~~Three sets of manufacturer's standard color guide for the specified coating system requiring a color selection.~~

PART 2 - PRODUCTS

2.01 GENERAL REQUIREMENTS

- A. For each painting and coating system, the required surface preparation, prime coat, intermediate coat (if required), topcoat, and coating thicknesses are specified herein. Mil thicknesses shown or specified are minimum dry-film thicknesses.

- B. Provide materials for a specified painting system, including primer, intermediate, and finish coats by the same manufacturer. Thinners, cleaners, driers, and other additives shall be as recommended by the paint manufacturer for the particular coating system.
- C. Deliver paints, whether for shop or field applications, in the original, unopened containers.

2.02 COATING SYSTEMS

- A. The following schedule is a guide to the types of surfaces requiring painting and is not intended to be complete or exact. Generally, all metal surfaces are to be painted for protection unless otherwise specified differently herein.
- B. System S-1: Ferrous metal subject to water immersion, excluding stainless steel and mating machined surfaces.
 - 1. Epoxy (or Zinc) Primer Application
 - a. Once surface preparation has been completed and accepted, coating application can begin.
 - b. Prior to primer spray application, all welds, rivets, edges, and corners shall be stripe coated by brush or roller. Ensure the coating is well worked in to the substrate.
 - c. Once stripe coating has been completed the spray coat can be applied.
 - d. Minimum DFT shall be 6.0 mils.
 - e. Products: Series 94H₂O Hydro-Zinc by Tnemec Inc.
 - 2. Midcoat
 - a. Once the primer coat has properly dried/cured and has been accepted, the midcoat application can begin.
 - b. Prior to spray application, all welds, rivets, edges, and corners shall be stripe coated by brush.
 - c. Ensure that coating is well worked in to substrate.
 - d. Once stripe coating has been completed the spray coat can be applied.
 - e. Minimum DFT shall be 6.0 mils
 - f. Products: Series 22 Epoxoline by Tnemec Inc.
 - 3. Topcoat
 - a. Once the midcoat has been completed and accepted, the topcoat shall be applied.
 - b. Minimum topcoat DFT shall be 6.0 mils.
 - c. The minimum total system DFT shall be 18.0 mils.
 - d. Products: Series 22 Epoxoline by Tnemec Inc.

D. Other Acceptable Paint Products

System	Manufacturer	Prime Coat	Intermediate Coat	Finish Coat
System S-1	Tnemec	Series 94H ₂ O Hydro-Zinc	Series 22 Epoxoline	Series 22 Epoxoline
	Carboline	Reactamine 760	Reactamine 760	Carbothane 133 HB
	Carboline	Carboguard 635	Carboquick 200	Carboquick 200
	PPG Industries	Sigmashield 1200	Sigmashield 1200	Sigmashield 1200

COATING SCHEDULE

No.	Equipment, Material, Surface or Substrate	System No.	Color of Final Coat
1	Stoplogs & Lifting Beam	S-1	Light Blue or Light Grey
2	General Field Touchup of Fabricator's Shop-Applied Coatings	*	Match Color

* Restore/touch-up shop-applied coatings in the field using touch-up paint furnished by the equipment fabricator.

PART 3 - EXECUTION

3.01 PROTECTION

- A. Remove, mask, or otherwise protect hardware, concrete surfaces, machinery, and other surfaces not intended to be painted.
- B. Provide drop cloths to prevent paint materials from falling on or marring adjacent surfaces.
- C. Remove and clean up paint drips, spills, and splashes from areas not being painted.

3.02 SURFACE PREPARATION

- A. Prepare all surfaces to be coated in accordance with the paint manufacturer's written specifications and instructions.
- B. Prepare steel surfaces for prime and finish coats in accordance with the Steel Structures Painting Council (SSPC) surface preparation specifications set forth below:

Specification	Type	Profile (Ave. Mils)
SP-1	Solvent Cleaning	--
SP-2	Hand Cleaning	--
SP-3	Power Tool Cleaning	--
SP-5	White Metal Blast Cleaning	1.5 – 2.5
SP-6	Commercial Blast Cleaning	1.5 – 3.0
SP-10	Near White Metal Blast Cleaning	1.5 – 3.0

- C. Do not sandblast or prepare more surface area in one day than can be coated in one day; prepare surfaces and apply coatings the same day. Remove all sharp edges, burrs, and weld spatter.
- D. Remove oil and grease from metal surfaces in accordance with SSPC SP-1. Use clean cloths and cleaning solvents and wipe dry with clean cloths. Do not leave a film or greasy residue on the cleaned surfaces before sandblasting.
- E. Remove weld spatter and weld slag from metal surfaces and grind smooth any rough welds, beads, peaked corners, sharp edges, erection lugs, etc. in accordance with SSPC SP-2 and SSPC SP-3.
- F. Neutralize welds with a chemical solvent that is compatible with the specified coating materials. Use clean cloths and chemical solvent. Wipe dry with clean cloths. Do not leave a residue on the cleaned surfaces.

ABRASIVE BLAST CLEANING

- A. Abrasive blast cleaning is defined as cleaning the surface through the use of very fine abrasives, such as siliceous or mineral abrasives, 80 to 100 mesh. Apply a fine etch to the metal surface to clean the surface of any contamination or oxide.
- B. Use dry abrasive blast cleaning for metal surfaces. Do not use abrasives that have become contaminated in automatic equipment. When shop or field blast cleaning with handheld nozzles, do not recycle or reuse blast particles.
- C. After blast cleaning and prior to application of coating, dry clean surfaces to be coated by dusting, sweeping, and vacuuming to remove residue from blasting.
- D. Keep the area of the work in a clean condition and do not permit blasting particles to accumulate and constitute a nuisance or hazard.
- E. During sandblast cleaning, prevent damage to adjacent coatings. Schedule blast cleaning and coating such that dust, dirt, blast particles, old coatings, rust, mill scale, etc., will not damage or fall upon wet or newly coated surfaces.
- F. Reclean prior to application of primer or touch-up coating any blast-cleaned surface not coated within said eight-hour period.

3.03 PAINT MIXING

- A. Prepare multiple-component coatings using all of the contents of the container for each component as packaged by the paint manufacturer. Do not use partial batches. Do not use multiple-component coatings that have been mixed beyond their pot life. Provide small quantity kits for touch-up painting and for painting other small areas. Mix only the components specified and furnished by the paint manufacturer. Do not intermix additional components for reasons of color or otherwise, even within the same generic type of coating. When power mixing multiple components, do not introduce moisture or air into product.

PRIME COATS

- A. Apply prime coats in accordance with the coating manufacturer's written specifications and instructions. Prime painting shall not be performed under adverse ambient conditions that may affect the life or appearance of the coating.
- B. Prime prepared surfaces of steel and iron immediately after completion of surface preparation.
- C. Do not apply prime coat over damp or moist surfaces.
- D. Prime coats found to be incompatible with finish coats or applied to improperly prepared surfaces shall be removed and an appropriate prime coat applied at no additional cost to the Agency.
- E. Apply primer immediately after blast cleaning and before any surface rusting occurs, or any dust, dirt, or any foreign matter has accumulated. Re-clean surfaces by blast cleaning that have surface colored or become moist prior to coating application.
- F. Apply a brush coat of primer on welds, sharp edges, nuts, bolts, and irregular surfaces prior to the application of the primer and finish coat. The brush coat shall be done prior to and in conjunction with the spray coat application. Apply the spray coat over the brush coat.
- G. Cure prime coats in accordance with the manufacturer's written specifications and instructions.

INTERMEDIATE COATS

- A. After proper application of the prime coat, apply by brush, one stripe coat of the intermediate coat epoxy, thinned 15%, to all welds, edges and corners, fasteners, and/or other difficult to reach areas.

FINISH COATS

- A. Apply finish coats in accordance with the coating manufacturer's written specifications and instructions. Follow the recommendations of the coating manufacturer including the selection of spray equipment, brushes, rollers, cleaners, thinners, mixing, drying time, temperature and humidity of application, and safety precautions. Painting shall not be performed under adverse ambient conditions that may affect the life or appearance of the coating.
- B. Finish coats found to be incompatible with prime coats or applied to improperly prepared surfaces shall be removed and an appropriate finish coat applied at no additional cost to the Agency.
- C. Do not apply coatings over damp or moist surfaces.
- D. Prepare undercoats and intermediate coats to differ noticeably in coloration for each other on succeeding coating applications to ensure even coverage by shading or tinting, as appropriate, for the paint material and color. Shading or tinting shall be done by the paint manufacturer.
- E. Conform to the requirements of SSPC PA-1, Shop, Field and Maintenance Painting.
- F. Stir, strain, and keep coating materials at a uniform consistency during application. Apply each coating evenly, free of brush marks, sags, runs, and other evidence of poor workmanship.

- G. Do not use thinners unless recommended by the coating manufacturer. If thinning is allowed, do not exceed the maximum allowable amount of thinner per gallon of coating material. Stir coating materials at all times when adding thinner. Do not flood the coating material surface with thinner prior to mixing. Do not reduce coating materials more than is absolutely necessary to obtain the proper application characteristics and to obtain the specified dry-film thicknesses.
- H. Remove dust, blast particles, and other debris from blast-cleaned surfaces by dusting, sweeping, and vacuuming. Allow ventilator fans to clean airborne dust to provide good visibility of working area prior to coating applications. Remove dust from coated surfaces by dusting, sweeping, and vacuuming prior to applying succeeding coats.
- I. Apply coating systems to the specified minimum dry-film thicknesses as measured from above the peaks of the surface profile.
- J. Finished surfaces shall be free from holidays, pinholes, bubbles, runs, drops, ridges, waves, laps, excessive or unsightly brush marks, and variations in color, texture, and gloss. Verify absence of holidays, pinholes, gaps, etc. by appropriate testing. Recoat and retest any areas failing these tests.
- K. Cure coats in accordance with the manufacturer's written specifications and instructions.

COATING INSPECTIONS

- A. Measure coating thickness specified for metal surfaces with a calibrated magnetic-type dry-film thickness gauge. Provide dry-film thickness gauge as manufactured by Mikrotest or Elcometer. Check each coat for the correct dry-film thickness. Do not measure within eight hours after application of the coating.
- B. A minimum of five spot measurements shall be taken in a given area to assure proper coat dry film thickness. The average of these measurements shall not be less than the specified thickness. No single spot measurement shall be less than 80 percent, or more than 120 percent, of the specified thickness.
- C. Test the coating with a low-voltage (22.5 to 80 volts with approximately 80,000 ohm resistance) holiday detector, using a sponge saturated with a 0.5 percent sodium chloride solution. The coating shall be holiday-free.
- D. Prepare and submit a report of prime, intermediate, and finish coatings inspections.

CONTACTING SURFACES

- A. When ordinary bolted contact is to exist between surfaces of ferrous or other metal parts of substantially similar chemical composition, such surface will not be required to be painted, but any resulting crevices shall subsequently be filled or sealed with paint.
- B. Contacting metal surfaces formed by high-strength bolts in friction-type connections shall not be painted.
- C. Where a nonmetal surface is to be in bolted contact with a metal surface, the contacting surfaces of the metal shall be cleaned and given three coats of the specified primer.

HANDLING AND TRANSPORTING SHOP-COATED ITEMS

- A. When loading for shipment to the project site, use spacers and other protective devices to separate items to prevent damaging the shop-coated surfaces during transit and unloading. If wood spacers are used, remove wood splinters and particles from the shop-primed surfaces after separation. Use padded chains or ribbon binders to secure the loaded items and minimize damage to the shop-coated surfaces.
- B. Handle shop-coated items with care during loading/unloading, installation, and erection operations to minimize damage. Do not place or store shop-coated items on the ground or on top of other work unless ground or work is covered with a protective covering or tarpaulin. Place shop-coated items above the ground upon platforms, skids, or other supports.
- C. Cover shop-coated items 100 percent with protective coverings or tarpaulins to prevent deposition of road salts, fuel residue, and other contaminants in transit.

FIELD PAINTING AND TOUCH-UP

- A. Clean and recoat surfaces with the specified paint material to obtain the specified color and coverage, if the surfaces have an improper finish color or insufficient film thickness.
- B. Sandblast or power-sand visible areas of chipped, peeled, or abraded paint, feathering of edges.
- C. Prime and finish coat in accordance with the paint manufacturer's written specifications and instructions. Work shall be free of runs, bridges, shiners, laps, or other imperfections.

UNCOATED SURFACES

- A. Do not paint the following surfaces unless otherwise directed by the Contracting Officer's Representative:
 - 1. Stainless steel.
 - 2. Plastic, Poly, and rubber.
 - 3. Bronze, aluminum, and nonferrous materials.

END OF SECTION

1.01 SUMMARY

- A. This section applies to the steel ~~stoplogs~~ and lifting beam.
- B. *BIA is adopting this section from the original fabrication specifications. Changes from the original specification are indicated herein by strikeout (for deletions) and italics (for additions).*"

1.02 REFERENCE SPECIFICATIONS, CODES, AND STANDARDS

A. American Institute of Steel Construction (AISC)

1. Code of Standard Practice for Steel Buildings and Bridges.
2. Specifications for ~~the Design, Fabrication, and Erection of Structural Steel for Buildings.~~

B. ASTM International (ASTM)

1. ~~AISC Manual of Steel Construction~~
2. ~~AISI Design of Light Gauge, Cold-Formed Steel Structural Members~~
3. ASTM A36 Carbon Structural Steel
4. ~~ASTM A48 Gray Iron Castings~~
5. ASTM A53 Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
6. ASTM A123 Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
7. ASTM A153 Zinc Coating (Hot-Dip) on Iron and Steel Hardware
8. ~~ASTM A193 Alloy Steel and Stainless Steel Bolting Materials for High Temperature Service~~
9. ~~ASTM A194 Carbon and Alloy Steel Nuts for Bolts for High Pressure and High Temperature Service~~
10. ASTM A307 Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength
11. *ASTM F3125 Standard Specification for High Strength Bolts* ~~ASTM A325 Structural Bolts, Steel, Heat Treated, 120/105 ksi~~
12. ~~ASTM A500 Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes~~
13. ASTM A992 Steel for Structural Shapes ~~for Use in Building Framing~~
14. ~~ASTM B21 Standard Specification for Naval Brass Rod, Bar, and Shapes~~
15. ~~ASTM D412 Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers—Tension~~
16. ~~ASTM D471 Standard Test Method for Rubber Property—Effect of Liquids~~
17. ~~ASTM D572 Standard Test Method for Rubber—Deterioration by Heat and Oxygen~~

~~18. ASTM D2137 Standard Test Method for Rubber Property—Brittleness Point of Flexible Polymers and Coated Fabrics~~

~~19. ASTM D2240 Standard Test Method for Rubber Property—Durometer Hardness~~

C. American Welding Society (AWS)

1. AWS D1.1 Structural Welding Code – Steel

~~2. AWS D1.6 Structural Welding Code—Stainless Steel~~

3. AWS QC1 Qualification and Certification of Welding Inspectors

D. ~~The Society for Protective Coatings~~

~~1. SP 1 Solvent Cleaning~~

~~2. SP 7 Brush-Off Blast Cleaning~~

1.03 CONTRACTOR SUBMITTALS

Submit the following in accordance with the requirements of Section 01 33 00 – Submittals:

~~All submittals shall be provided in electronic format and include the following:~~

A. The contractor shall prepare and submit shop drawings for the steel stoplogs and the lifting beam. The dimensions and reference materials of construction by ASTM designation and grade shall be shown in the drawings. Also include complete fabrication and weld details, tolerances and finish requirements.

1. The weights of all major items shall be listed.

2. Fabrication schedule.

B. Shop inspection and test plan.

1. Weld procedure specifications, procedure qualification records including all destructive and nondestructive test results and welding bead profiles as required along with individual welder qualification certificates.

C. Certified copies of material test reports.

D. Manufacturer's data sheet for each product proposed, including statements on the suitability of the material for the intended use.

~~E. Paint manufacturer's color chart, instructions and recommendations on surface preparation and application.~~

~~F. Material Safety Data Sheet (MSDS) for each product proposed.~~

G. Quality control records of all material test ~~reports~~ records, shop inspection and testing, dimensional checks and test data.

1.04 QUALITY ASSURANCE

A. Establish and maintain quality control for the Work specified to ensure compliance with the Contract Documents during fabrication.

- B. Fabricate structural steel members in accordance with the AISC Code of Standard Practice *and additional tolerances indicated on design drawings, if any.*
- C. Perform welding with electric arc process and in accordance with AWS D1.1, "Structural Welding Code".
- D. In addition to complying with pertinent codes and regulations, comply with:
 - 1. AISC "Code of Standard Practice for Steel Buildings and Bridges".
 - 2. AISC "Specifications for ~~the Design, Fabrication, and Erection of~~ Structural Steel Buildings" and including the "Commentary on the AISC Specification".

1.05 QUALITY RECORDS

- A. Inspection of all materials to be used for fabrication before *starting* fabrication.
- B. Maintenance of all quality control records during fabrication.
- C. Maintenance of all records of shop inspections, including dimensional tolerances, finishes, mating of interfacing parts, and non-destructive testing of welds.

1.06 MAXIMUM FABRICATION DURATION

- A. *The Contractor may determine and schedule the pickup date of the equipment at its discretion. However, once the equipment is picked up, the Contractor shall comply with the following schedule requirements:*
 - 1. *21-Day Modification Period: Beginning on the date of pickup, Contractor has 21 days to modify the equipment and prepare it for the Owner's shop inspection.*
 - 2. *Scheduling the Shop Inspection: Contractor will comply with the advance notice requirement specified in this document. Unless authorized or directed by the Owner's Representative, the Contractor shall schedule the shop inspection to occur not later than three (3) days after the end of the Contractor's 21-day modification period.*
 - 3. *Post Inspection Adjustments and Delivery: Following the shop inspection, the Contractor has five (5) days to make any required post-inspection adjustments and deliver the equipment to the Owner.*

2.01 GENERAL

- A. All materials incorporated in the *modification* work shall be new and both workmanship and materials shall be of first-class quality, free from defects and imperfections, and, where indicated on the Plans, of the classification and grades designated therein.
- B. Material standards including grade, class, or type shall be shown on shop drawings prepared by Contractor. If equivalent material standards are to be used, complete details of the standards shall be submitted for review and approval by the BIA Contracting Officer's Representative, including a list of the materials, their ASTM equivalents, and the identification of components of equipment for which they are to be used.

- C. Defective material shall not be repaired and used in the construction of the equipment without prior approval. No peening, caulking, or filling shall be permitted in repairing cracks, pinholes, or blowholes. Defects in welds shall be repaired by chipping out to sound metal and re-welding. Repair materials in accordance with ASTM or similar standards to the extent and in the manner permitted by the standards.

2.02 MATERIALS

A. Steel

Table 2.1

Wide Flange Shapes	ASTM A992
Shapes, Plates, Bars	ASTM A36
Pipe, Pipe Columns, Bollards	ASTM A53, Type E or S, Grade B standard weight unless indicated otherwise
HSS	ASTM A500 Grade B

B. Rubber Seals

1. ~~The rubber water seals shall be the product of a reputable seal supplier such as Seals Unlimited, 6410 NE Jacobson St., Hillsboro, OR 97124 (information available online at: www.sealsunlimited.com) or approved equal.~~
2. ~~The seals shown on the drawings shall be neoprene or EPDM compounded and cured to provide the physical properties listed in Table 2.2~~

~~Tests to determine physical properties of the seals material shall be conducted in accordance with applicable provisions of the test methods listed in Table 2.2. Specimens for seal tests shall be prepared from each batch of compound mixed for use in fabricating the seals to be furnished.~~

Table 2.2 — Physical Properties of Seals

Physical Property	ASTM Test Method	Quantity
Minimum Ultimate Tensile Strength (lb/in ²)	D 412, Method A	2,500
Minimum Elongation at Break (Percent)	D 412	450
Shore A Durometer Hardness (Unitless)	D 2240	60 to 70
Maximum Absorption of Water (70 hours @ 100° C; Percent Change in Volume)	D 471	10
Maximum Compression Set (Percent of Original Test Deflection)	D 395, Method B	30
Minimum Tensile Strength after Oxygen Bomb Aging (Percent of Ultimate Tensile Strength Before Aging)	D 572	80
Low Temperature Brittleness	D 572	Non-brittle after 3 minutes @ -40° C
Maximum Change in Weight (46 Hours in Water at 70° C; Percent)	D 471	5

3. ~~Seal strips shall be molded or extruded in convenient lengths and, if necessary, shall be butt spliced together by vulcanizing to obtain the minimum continuous lengths shown on the drawings. The number of splices in a required minimum continuous length of seal strip, not including splices with corner seals, shall not exceed one for each 9 feet of length and the minimum length of seal strip for splicing shall not be less than 2 feet.~~
4. ~~All seal splices shall be vulcanized in molds using heat and pressure. Wrapped splices are not acceptable. Each splice shall have sufficient strength to allow 50 percent elongation over an element of seal strip which includes the splice. All splicing shall be performed by the seal manufacturer. To demonstrate the integrity of the splice process, the seal manufacturer shall perform at least two elongation tests of a sample seal splice in accordance with test method of ASTM designation: D 412. The sample splice joint shall be prepared with elements of seal strips made from the same batch compound mixed for use in fabricating the seals to be furnished. The results of these elongation tests shall be made available to the Government, upon request. Between initial fabrication of seal splices and final acceptance of installed seals by the Government, any splices that fail for show deficiencies (e.g., separation, holes, incomplete bonding, misalignment, etc.) shall be re-spliced.~~

~~At vulcanized splices, the dimensions of the cross sections to be joined shall be matched to avoid any pronounced offsets that might result in leakage. For joints in music note seals, the cylindrical sealing surface of the bulb and the face of the stem that is tangent to the bulb shall be flush. Offsets to these surfaces at all spliced joints shall be buffed flush with a transition slope not exceeding 1/16 inch in 24 inches of seal length.~~

~~The splices that are not designated to be vulcanized shall be cemented together at shop installation, by the stoplog fabricator. At final assembly, seal joint surfaces shall be buffed with 80 grit sandpaper and joined together with cement recommended by seal manufacturer.~~

5. ~~Tolerances for the cross section dimensions of music note seals are shown on drawings 40-D-6274 and 40-D-6275.~~
6. ~~The contractor shall be responsible for all costs and time delays resulting from defective seals, including any additional testing required by the Government to ensure that sound and durable seals have been obtained.~~
7. ~~Corner seals shown on drawing 1117-D-589 shall be molded in one piece or vulcanized as shown, with sufficient excess length to allow for cutting and fitting to the joint locations dimensioned on the drawings.~~
8. ~~The seals shall be fabricated and cured in such a manner that any cross section will be dense, homogeneous, and free from porosity and other imperfections. Exterior surfaces of the seals shall be free from pits, blisters, and other imperfections that will impair sealing. Dimensions shall be uniform from one seal cross section to the next. Variations in dimensions between cross section of continuous seal strips shall not exceed a rate of change of 1/16 inch in 24 inches of seal length, except that measurements of cross sections should not be checked at intervals less than 6 inches apart.~~
9. ~~Unless otherwise specified, all hardware for new rubber seals shall be of stainless steel ASTM A276, type 304/316 with stainless steel plain washer and neoprene seal washer.~~

~~The nuts used with stainless steel bolts shall be of bronze per ASTM B 21, Alloy No. 464 half hard (naval brass).~~

C. ~~Stainless Steel~~

- ~~1. Unless otherwise indicated, stainless steel metalwork and bolts shall be fabricated from Type 304 stainless steel.~~

D. *Ultra-High Molecular Weight Polyethylene (UHMW)*

1. *UHMW material shall be in accordance with ASTM D4020; virgin, extruded or molded, white*

2.03 STRUCTURAL STEEL FABRICATION

- A. Design and fabrication of structural parts shall conform to the applicable provisions of the AISC Specifications for the ~~Design, Fabrication, and Erection of Structural Steel for Buildings, and of the AISC~~, Code of Standard Practice for Steel Buildings and Bridges.
- B. Fabrication and assembly shall be performed in the shop.
- C. All bolts, nuts, screws, rivets, threads, pipe, and measurements or dimensions shown on the Plans shall conform to English standards.
- D. Manufacturing Process. The Contractor shall be responsible for the proper selection of manufacturing methods and assembly. The Contractor shall also be responsible for the correct fit of parts.
- E. Workmanship. All work shall be performed and completed in a thorough workmanlike manner and shall follow the best modern practices in the design and manufacture of heavy welded structures. All work shall be done by personnel skilled in the related professions and operations. All parts shall be made accurately to standard gages so as to facilitate replacement and repairs. All special gages and templates necessary for field erection shall become the property of the Government. Like parts shall be interchangeable wherever possible. Machining fits on renewable parts shall be accurate and to specific dimensions so that replacements made to the size shown on the shop drawings may be readily installed.
- F. Structural Steel Fabrication and Erection. Material must be straight before being laid off or worked. If straightening is necessary, it shall be done by methods that will not impair the metal. Sharp kinks or bends shall be cause for rejection of the material. Material with welds will not be accepted except where welding is definitely specified, indicated or otherwise approved. Bends shall be made by approved dies, press brakes or bending rolls. Where heating is required, precautions shall be taken to avoid overheating the metal and it shall be allowed to cool in a manner that will not impair the original properties of the metal. Proposed flame cutting of material other than structural steel shall be subject to approval of the BIA Contracting Officer's Representative and shall be indicated on shop drawings. Shearing shall be accurate and all portions of the work shall be neatly finished. Corners shall be square and true unless otherwise shown on the Plans. Re-entrant cuts shall be filleted to a minimum radius of 3/4-inch unless otherwise approved by the BIA Contracting Officer's Representative. Finished members shall be free of twists, bends and open joints. Steel may be cut by mechanically guided or hand guided torches, provided an accurate profile with a surface that is smooth and free from cracks and notches is obtained. Surfaces and edges to be welded shall be prepared in accordance with

AWS D1.1. Where structural steel is not to be welded, chipping or grinding will not be required except as necessary to remove slag and sharp edges of mechanically guided or hand guided cuts not exposed to view. Hand guided cuts which are to be exposed or visible shall be chipped, ground or machined to sound metal.

- G. Dimensional Tolerances for Structural Work. Dimensions shall be measured using calibrated steel tapes and measuring devices of approximately the same temperature as the material being measured. The overall dimensions of an assembled structural unit shall be within the tolerances indicated on the drawings, as specified in the particular section of the contract documents for the item of work, or as required for the proper operation of the equipment. Where tolerances are not specified an allowable variation of 1/32-inch is permissible in the overall length of component members with both ends milled. Component members without milled ends shall not deviate from the dimensions shown by not more than 1/16 inch for members 30 feet or less in length and by more than 1/8-inch for members over 30 feet in length. Erection tolerances not specified shall be as noted in AISC S303, Code of Standard Practices for Steel Buildings and Bridges, ~~Section 7.~~

2.04 SHOP WELDING

- A. Codes. All welding shall be welded in accordance with AWS D.1 or D1.6, as applicable. The Contractor shall be responsible for developing the procedure and performing the welding in accordance with the approved procedure at no additional cost to the Government.
- B. General. All welds shall be continuous and watertight. All welding shall be performed by the electric-arc method, by a process that excludes the atmosphere from the molten metal, and where practicable, by automatic machines. Machined surfaces of parts affected by welding shall be machined to final dimensions after welding. Machined surfaces of parts requiring stress relief shall be machined to final dimensions after the parts have been stress relieved. Localized stress relieving shall not be permitted on shop welded parts. Welder qualification, weld procedure qualification, base metal preparation, preparation for welding, weld electrode selection, weld finish, inspection and acceptance criteria and methods used in correcting defective welds shall be per AWS D1.1. Certificates of welders' qualifications shall be submitted when requested. Weld procedures shall be submitted for approval by the Engineer.
- C. Minimum Weld Requirements. Unless noted otherwise in the contract documents, the minimum size of fillet welds shall be 1/4-inch measured on the leg. Welds larger than 5/16-inch shall not be made in less than two passes. All groove welds including butt welds shall be full penetration, welded from both sides.
- D. Filler Metal. The electrode and electrode-flux combination shall conform to the appropriate AWS specification for the base metal and welding process being used and shall not have a minimum tensile stress less than the strength of the base or parent metal. The AWS designation of the electrodes to be used shall be included in the schedule of welding procedures. Only low hydrogen electrodes shall be used for manual shielded metal-arc welding. A controlled temperature storage oven shall be used at the job site as to maintain low moisture of low hydrogen electrodes.
- E. Welding of Splices. Splice welds shall be made using AISC approved pre-qualified full penetration welds, ground flush after welding.

- F. Weld Finish. Welds shall in general be treated so that they will display good appearance and a surface suitable for painting. Welds shall be ground and blended to eliminate stress risers. All welds that require nondestructive examinations shall be dressed by chipping and grinding as required for good interpretation by the selected weld examination method. Final grinding marks shall be perpendicular to the weld axis.
- G. Repair of Welds. Defects in welds shall be repaired and re-tested per AWS D1.1. Defective welds shall be repaired and re-inspected at no additional cost to the Government.
- H. Stress Relief Heat Treatment. Where stress relief heat treatment is specified or shown on the Plans it shall be in accordance with the requirements of AWS D1.1 unless otherwise authorized or directed by the Engineer.
- I. Inspector Qualification. All shop and field welding inspectors shall be qualified in accordance with the requirements of AWS QC1 and nondestructive testing personnel shall be qualified in accordance with the requirements of ASNT 01 for Levels I or II in the applicable nondestructive testing method. The inspector may be supported by assistant welding inspectors who are not qualified to ASNT 01, and assistant inspectors may perform specific inspection functions under the direct supervision of the qualified inspector.
- J. Nondestructive Testing of Welds
1. General. The Contractor shall maintain an approved inspection system and perform required inspections in accordance with the Contractor's approved welding procedures.
 2. Examination of Welds. Examination of Welds shall be as follows:
 - a. All (100%) completed welds shall be cleaned and carefully examined for insufficient throat or leg sizes, cracks, undercutting, overlap, excessive convexity ~~or reinforcement~~ and other surface defects.
 - b. All fillet welds shall be given a complete (100%) non-destructive examination by liquid penetrant (PT) methods, as applicable.
 - c. All partial penetration welds shall be tested by PT method.
 - d. All full penetration welds other than butt welds shall be given non-destructive examination throughout their entire length by ultrasonic examination (UT), followed by radiographic examination (RT) if UT results are not clear, or by PT if RT is not feasible.
 - e. Butt welds shall be examined by UT or PT if UT is not feasible.
 3. Acceptance Criteria. All welds shall comply with the requirements of AWS D1.1.
 4. All welds found to be defective shall be repaired prior to stress relieving. Where stress relieving is required, non-destructive examination shall be performed after stress relieving.
 5. Non-Destructive Testing Agency. The nondestructive examination of welds and the evaluation of examination tests as to the acceptability of the welds shall be performed by a testing agency or by the contractor using suitable equipment and qualified personnel. Personnel performing nondestructive testing shall be qualified and certified in accordance with the requirements of AWS QC-1-88. Non-destructive testing personnel

shall be qualified Level II inspectors as qualified under a certified program meeting the guidelines of the American Society of Non-Destructive Testing (ANST) standard SNT-TC-1A. The company performing non-destructive testing shall have employed a Level III inspector as qualified under a certified program meeting the guidelines of SNT-TC-1A to supervise the Level II inspectors. If requested by the Government, the Contractor shall submit, 14 days prior to beginning of testing, sufficient documentation demonstrating that the testing personnel are qualified.

6. Non-Destructive Testing Documentation. The Contractor shall maintain documentation of all visual and nondestructive weld testing. The documentation shall show that applicable sections of the specifications are met. Ultrasonic equipment qualification and weld inspection forms that document ultrasonic weld inspection shall be included. All non-destructive testing results shall be submitted for information and shall become the property of the Government. The documentation shall include corrective action taken at defective welds.

2.05 BOLTS

- A. Bolts used to connect dissimilar metals shall be ASTM A193 and A194, Type 316 stainless steel.
- C. All steel structural connection bolts shall be heavy hex-head conforming to ASTM *F3125 Grade A325, hot-dip galvanized finish*, unless otherwise noted.
- D. Nuts and washers shall conform to ASTM A563 and F436, respectively. ~~Provide self-locking nuts or lockwashers and plain nuts where shown on the plans.~~
- F. ~~Stainless steel bolts shall be ASTM A193, Grade B8M, Class 2, or ASTM F593, Type 316, Condition SH. Nuts shall be ASTM A194, Grade 8M, or ASTM F594, Type 316, Condition SH. Use ASTM A193 bolts with ASTM A194 nuts; use ASTM F593 bolts with ASTM F594 nuts. Provide a washer under each nut and under each bolt head. Washers shall be of the same material as the nuts.~~
- G. Certify that the bolts and nuts have been proof tested.
- H. Bolts and nuts shall be installed with washers fabricated from material matching the base material of bolts, except that hardened washers for high-strength bolts shall conform to the requirements of the AISC Specification.
- I. The length of each bolt shall be such that the bolt extends at least 1/8 inch beyond the outside face of the nut before tightening.
- K. Coating
 1. ~~Threads on stainless steel bolts shall be protected with an antiseize lubricant suitable for submerged stainless steel bolts, meeting government specification MIL-A-907E.~~
 2. Antiseize lubricant shall be classified as acceptable for potable water use by the NSF.
- L. ~~Antiseize lubricant shall be "PURE WHITE" by Anti-Seize Technology, Franklin Park, IL, 60131, AS-470 by Dixon Ticonderoga Company, Lakehurst, NJ, 08733, or equivalent.~~

2.06 BEVELED WASHERS

- A. Beveled washers for W-shapes, structural tees, channels and hollow structural sections shall be square or rectangular, tapered in thickness, smooth, and conforming to ASTM F436.

2.07 SHOP ASSEMBLY

- A. The stoplogs and lifting beam shall be completely assembled in the shop to ensure that all parts fit and that the required dimensions and tolerances have been obtained.
- B. ~~The seals shall be fitted and installed in the stoplog fabricators shop. To ensure water tightness; all splices between seal strips and corner seals, except as noted on the drawings, shall be vulcanized in molds using heat and pressure. Vulcanized splices shall conform to the requirements of Section 2.02, Subsection B, Part 4 for joining seal strips into continuous lengths and shall be performed by the seal manufacturer.~~
- C. ~~Seal splices that are not designated to be vulcanized shall be cemented together at shop installation by the stoplog fabricator. At final assembly, seal joint surfaces shall be buffed with 80-grit sandpaper and joined together with cement recommended by seal manufacturer.~~
- D. ~~Measure seal lengths accurately and cut seal ends at right angles to ensure tight fits before splicing joints. To cut seals, provide appropriate miter box that will hold and confine seals securely and use saw with no tooth set. Use seal clamps to locate and mark holes. Before marking and drilling holes, position the seals such that the required seal projection shown on the drawing will be obtained. Drill holes 1/8 inch oversize to allow for ease of assembly and adjustment of seals. Drill holes with a rotary plug drill, using ample water lubrication and soft wood or rubber backing.~~
- E. Before painting, each completely assembled stoplog section, including seal clamp bars and seals shall be balanced. Each stoplog section, while suspended from the lifting beam, shall be plumbed by permanently attaching metal weights in the areas shown on the drawings, so that the machined faces of the side seat bars are vertical within plus or minus one-fourth inch across the width of the stoplog.
- F. ~~Each stoplog set and the lifting beam shall be tested for proper engagement, alignment, interchangeability as follows:~~
- ~~1. Attach the lifting beam to each log section and check for engagement of hooks and dowels.~~
 - ~~2. Engage each log section with each other log section within their set and check alignment of seat bar faces. Dowels in dowel holes shall not interfere with alignment of seat bar faces. Shims may be required between the top and bottom angles on the upstream side to compensate for the bottom seal projection on the downstream side. The stoplogs shall also be checked for continuous contact of the bottom stoplog seals with the mating stoplog.~~
- G. ~~Final operation and correction of any deficiencies shall be approved by the Contracting Officer.~~
- H. *Contractor is responsible for rectifying nonconformities and delivering the modified equipment in accordance with the design drawings and specifications.*

2.08 BIA REPRESENTATIVE'S INSPECTION OF MATERIALS AND AUTHORIZATION FOR SHIPMENT

- A. *When fabrication is complete, the BIA Representative will inspect the materials specified under this section at the Contractor's fabrication shop prior to shipment.*
- B. *Contractor shall provide the BIA Representative with not less than seven (7) days' advance notice of the shop inspection date.*
- C. *Contractor shall provide access to the completed equipment at Contractor's fabrication shop for BIA Representative, and/or BIA Representative's testing personnel.*
- D. *Following their inspection, BIA Representative will either:*
 - 1. *Provide authorization to the Contractor to proceed with shipment, or*
 - 2. *Provide a list of nonconformities requiring correction prior to shipment.*
- E. *BIA Representative may elect to complete additional inspections at the Contractor's shop or at the point of delivery.*
- F. *Contractor shall not ship materials until the BIA Representative has completed their shop inspection(s) and authorized shipment.*

2.09 SHIPPING

A. *Pickup*

- 1. *The Contractor is responsible for transporting the lifting beam from the Headgate Rock Dam project site near Parker, Arizona, to the Contractor's shop.*
- 2. *BIA will provide equipment to assist with loading the lifting beam onto the Contractor's transport vehicle at the project site during pickup.*
- 3. *The Contractor shall coordinate with the BIA Representative in advance of pickup to schedule access and loading equipment.*

B. *Delivery*

- 1. *Following BIA Representative's authorization for shipment, the Contractor will deliver the modified lifting beam to BIA at the project site.*
- 2. *BIA will provide equipment to assist with unloading the lifting beam from the Contractor's transport vehicle at the project site during delivery.*
- 3. *The Contractor shall coordinate with the BIA Representative in advance of delivery to schedule access and loading equipment.*
- 4. *The Contractor must deliver the modified lifting beam to the site within the Maximum Fabrication Duration defined in Paragraph 1.06 of this specification.*

~~After completion of the shop assembly and inspection, the stoplogs and stoplog lifting beam shall be disassembled as necessary for shipment.~~

~~The seals, if shipped assembled on the stoplogs, shall be protected against any damage, including compression set. Seals that are not shipped assembled on the stoplogs shall be match-marked for field assembly to their respective positions.~~

- C. ~~Any wooden blocks, pallets, or other packing and stacking materials shall be retained for storage of the stoplogs onsite.~~

3.00 FIELD REPAIRS

- A. Field touch-up of damaged painting shall be performed in accordance with Section 05 05 133 - Painting and Coatings.

4.00 FINAL TESTING

~~A final test of the stoplogs will be performed once the stoplogs arrive onsite. Placement of the stoplogs in the stoplog slots will be accomplished by a BIA crane operator using the mobile crane located onsite. A successful test, as determined by a designated BIA representative, will conclude the project. The correction of any deficiencies will be approved by the Contracting Officer.~~

- A. *BIA will lead a Final Test at the Main Canal Headworks at Headgate Rock Dam. The Final Test will consist of lifting a stoplog using the modified lifting beam, then placing and withdrawing the lifting beam and stoplog combination through the north, upstream stoplog slots on the Main Canal Headworks to ensure the modified lifting beam travels freely, without binding. If there is too much gap between wall guides and wall surfaces or binding occurs, as determined by the BIA representative, the Contractor will adjust the wall guide position accordingly and the test will be repeated until satisfactory performance is demonstrated.*

The purpose of this test is to confirm that the end guide modifications provide an acceptable level of stoplog placement and withdrawal operability. The BIA Representative shall approve the date and time of the Final Test and witness the test.

B. *Equipment*

1. The Contractor shall provide a crane and crane operator to support BIA's field-directed testing.
2. The Owner will provide equipment for loading, unloading, and/or transporting the lifting beam from the storage area to the crane location.

- C. *A Contractor's representative must be present at the Final Test and will demonstrate the adjustable end guide functionality of the modified lifting beam as required.*

- D. *A successful test, as determined by a designated BIA representative, will conclude the project.*

E. *In the event of an unsuccessful Final Test:*

1. *If the Final Test reveals nonconformities between the modified equipment and the design drawings or specifications, Contractor shall promptly rectify the nonconformities and reschedule/repeat the Final Test.*
2. *The correction of deficiencies other than nonconformities identified during the Final Testing requires the approval of the Contracting Officer.*

END OF SECTION